REGULARITY OF HENSELIAN LOCAL RINGS ACROSS THE COMPLETION BASE CHANGE, WITH PROPER GLOBAL DESCENT

THE STACKS PROJECT AUTHORS

Editorial note. The closedness of |Y| over |Spec(A^)| used to find the specialization comes from properness of Y over A in the case (3) hypothesis.

Allowed standard prerequisites (editorial).

- Porting regularity to the completion for schemes. Resolution of Surfaces, lemma port-regularity-to-completion (cross-chapter, named). Scheme case of statements (1) and (2), imported by etale localization.
- Elementary etale neighbourhoods and specialization lifting. Decent Spaces, lemmas decent-space-elementary-etale-neighbourhood and decent-specialization (cross-chapter, named). Provide (V, v_0) -> (Y, y_0) and lift the specialization y -> y_0 into V.

Source reference. Original reference 0BBP: focused-seeds/geometry/S9-spres-port-regularity-to-completion/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IL: focused-seeds/geometry/S9-spres-port-regularity-to-completion/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BHQ: focused-seeds/geometry/S9-spres-port-regularity-to-completion/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local ring. Let $X \to \mathrm{Spec}(A)$ be a morphism which is locally of finite type with X a decent algebraic space. Set $Y = X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A^\wedge)$. Let $y \in |Y|$ with image $x \in |X|$. Then*

1. *if $\mathcal{O}^h_{Y,y}$ is regular, then $\mathcal{O}^h_{X,x}$ is regular,*
2. *if y is in the closed fibre, then $\mathcal{O}^h_{Y,y}$ is regular $\Leftrightarrow$ $\mathcal{O}^h_{X,x}$ is regular, and*
3. *If X is proper over A, then X is regular if and only if Y is regular.*

Proof. By étale localization the first two statements follow immediately from the counter part to this lemma for schemes, see Resolution of Surfaces, Lemma 0BHQ.

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For part (3), since $Y \to X$ is surjective (as $A \to A^\wedge$ is faithfully flat) we see that Y regular implies X regular by part (1). Conversely, if X is regular, then the henselian local rings of Y are regular for all points of the special fibre. Let $y \in |Y|$ be a general point. Since $|Y| \to |\operatorname{Spec}(A^\wedge)|$ is closed in the proper case, we can find a specialization $y \rightsquigarrow y_0$ with y_0 in the closed fibre. Choose an elementary étale neighbourhood $(V, v_0) \to (Y, y_0)$ as in Decent Spaces, Lemma 0BBP. Since Y is decent we can lift $y \rightsquigarrow y_0$ to a specialization $v \rightsquigarrow v_0$ in V (Decent Spaces, Lemma 03IL). Then we conclude that $\mathcal{O}_{V,v}$ is a localization of $\mathcal{O}_{V,v_0}$ hence regular and the proof is complete. $\square$